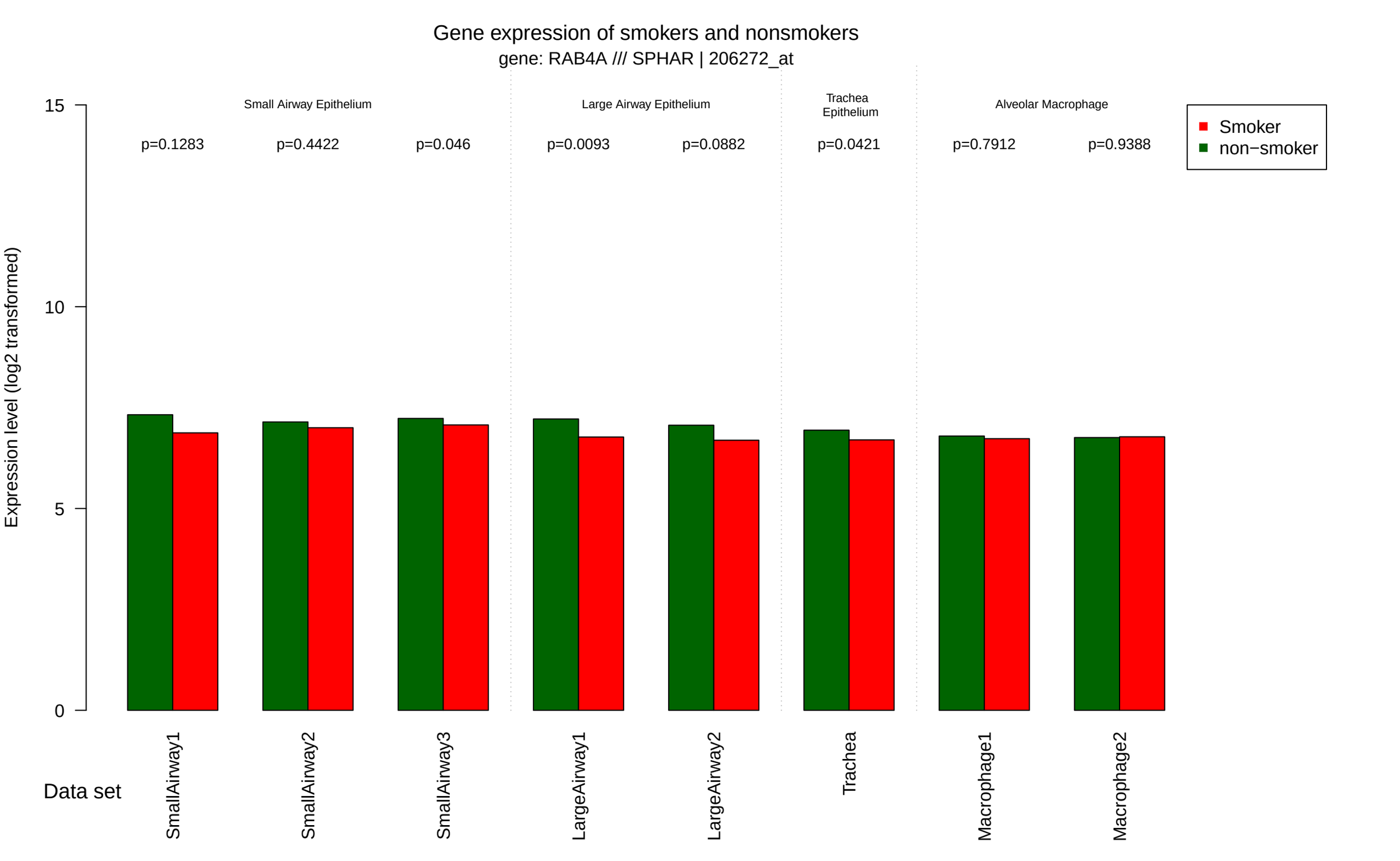
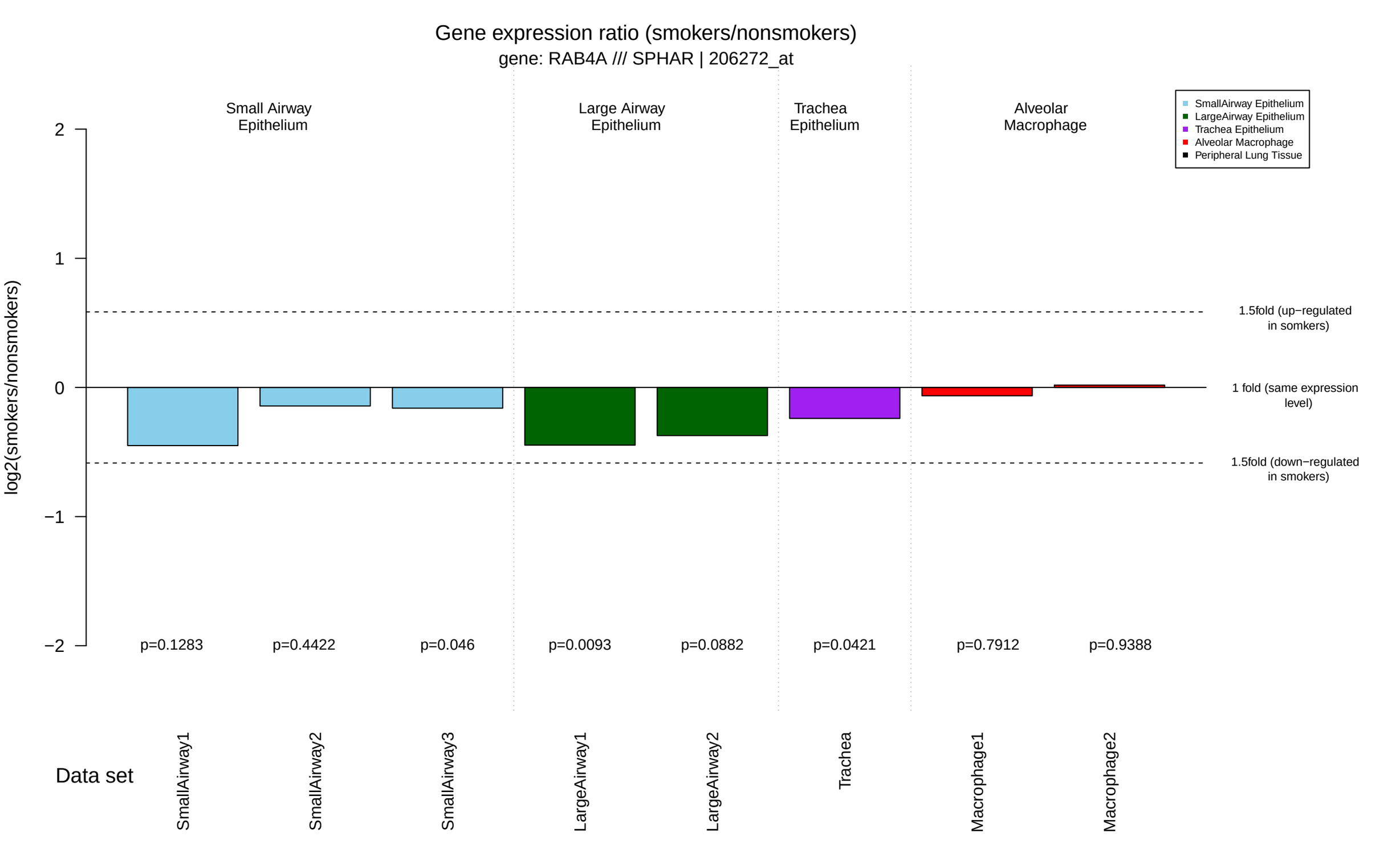
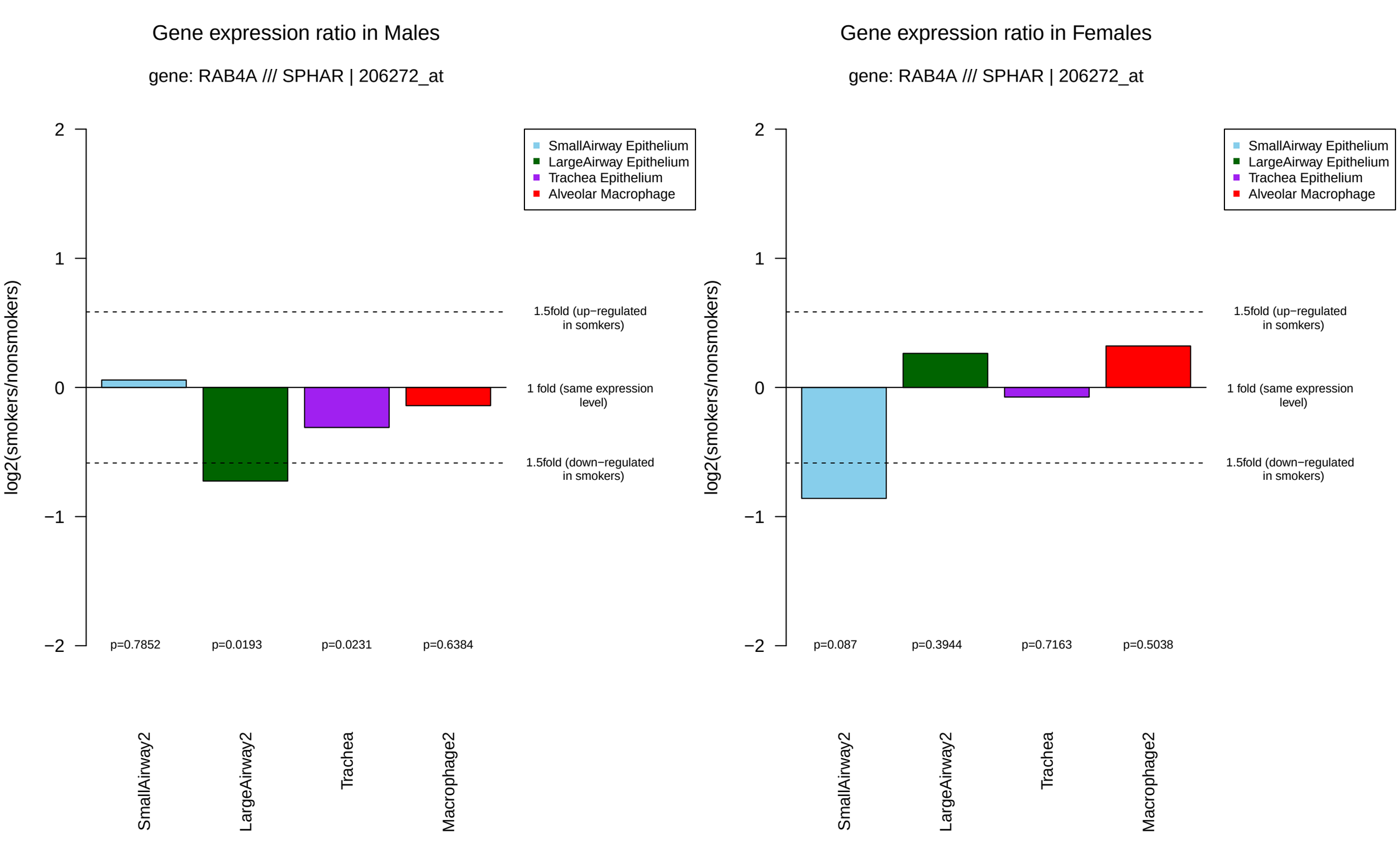
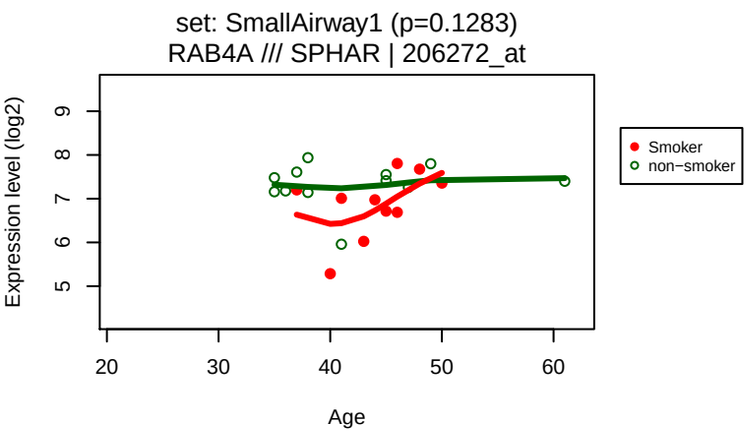
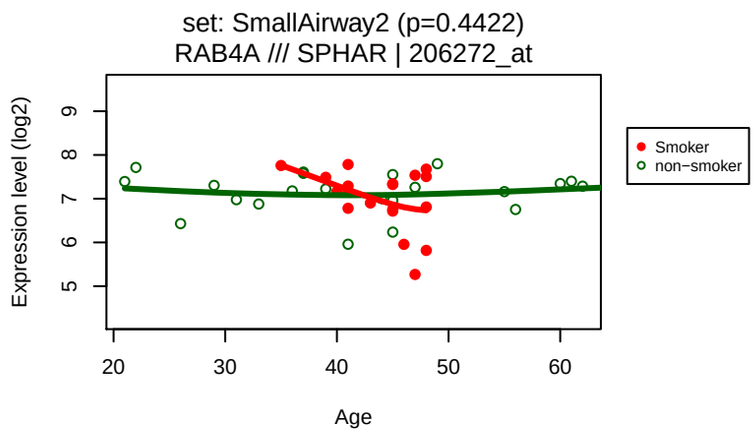
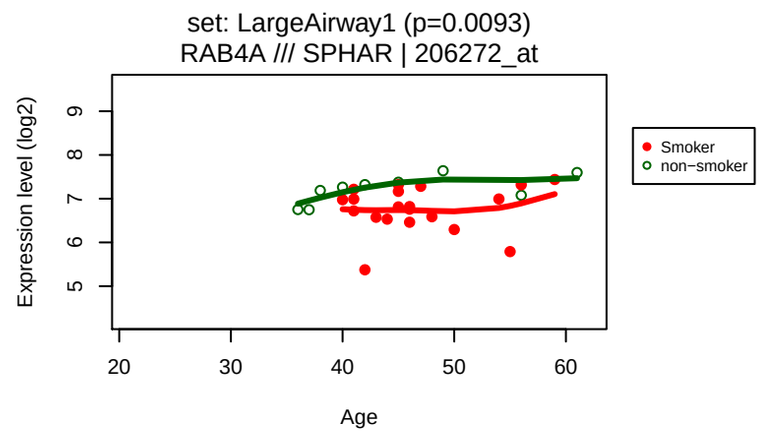

Gene expression of smokers and nonsmokers
gene: RAB4A /// SPHAR | 206272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1283
p=0.4422
p=0.046
p=0.0093
p=0.0882
p=0.0421
p=0.7912
p=0.9388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

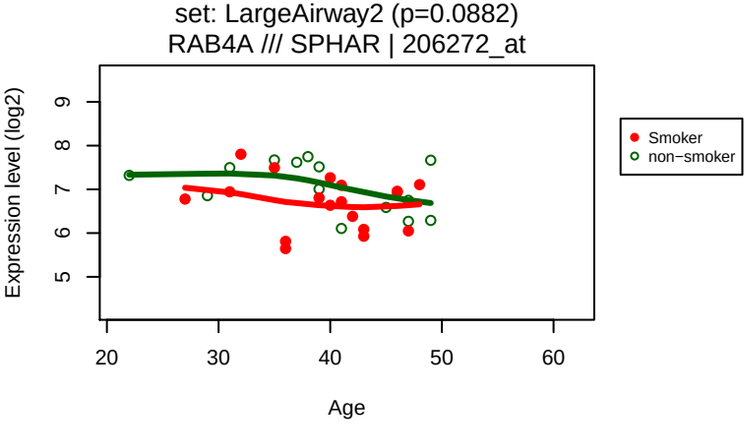

Gene expression ratio (smokers/nonsmokers)
gene: RAB4A /// SPHAR | 206272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1283
p=0.4422
p=0.046
p=0.0093
p=0.0882
p=0.0421
p=0.7912
p=0.9388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

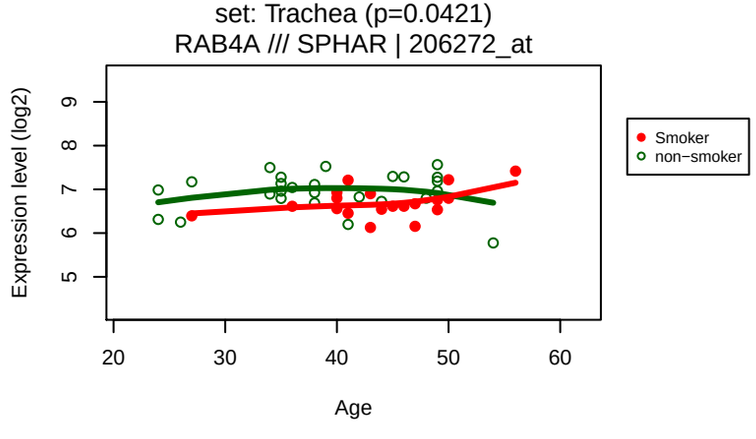

Gene expression ratio in Males
gene: RAB4A /// SPHAR | 206272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7852
p=0.0193
p=0.0231
p=0.6384
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB4A /// SPHAR | 206272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.087
p=0.3944
p=0.7163
p=0.5038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

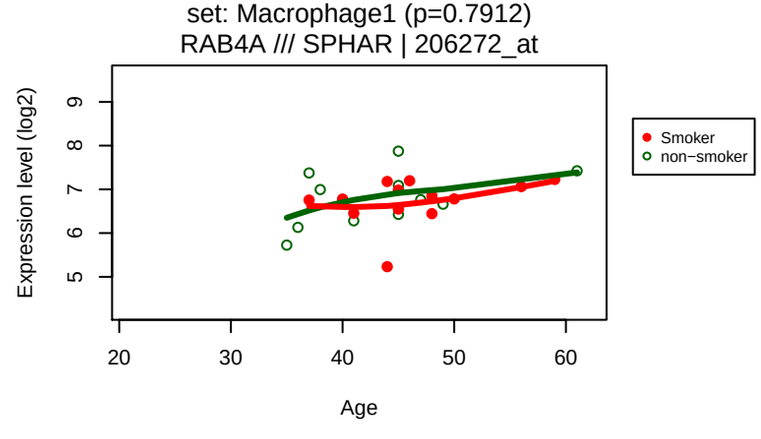

set: SmallAirway1 (p=0.1283)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker

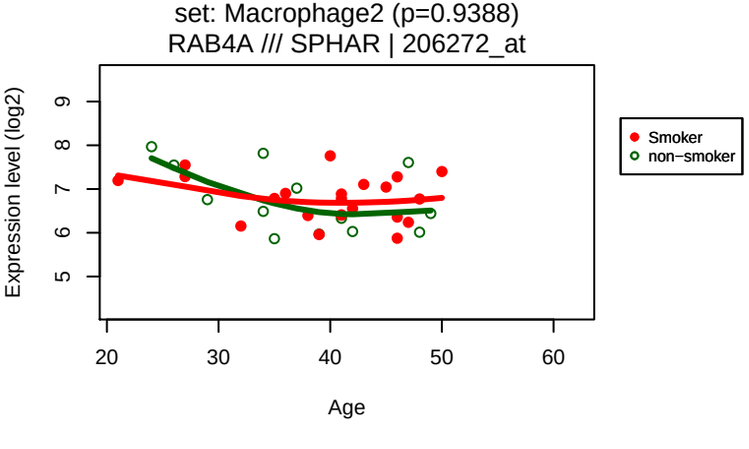

set: SmallAirway2 (p=0.4422)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker

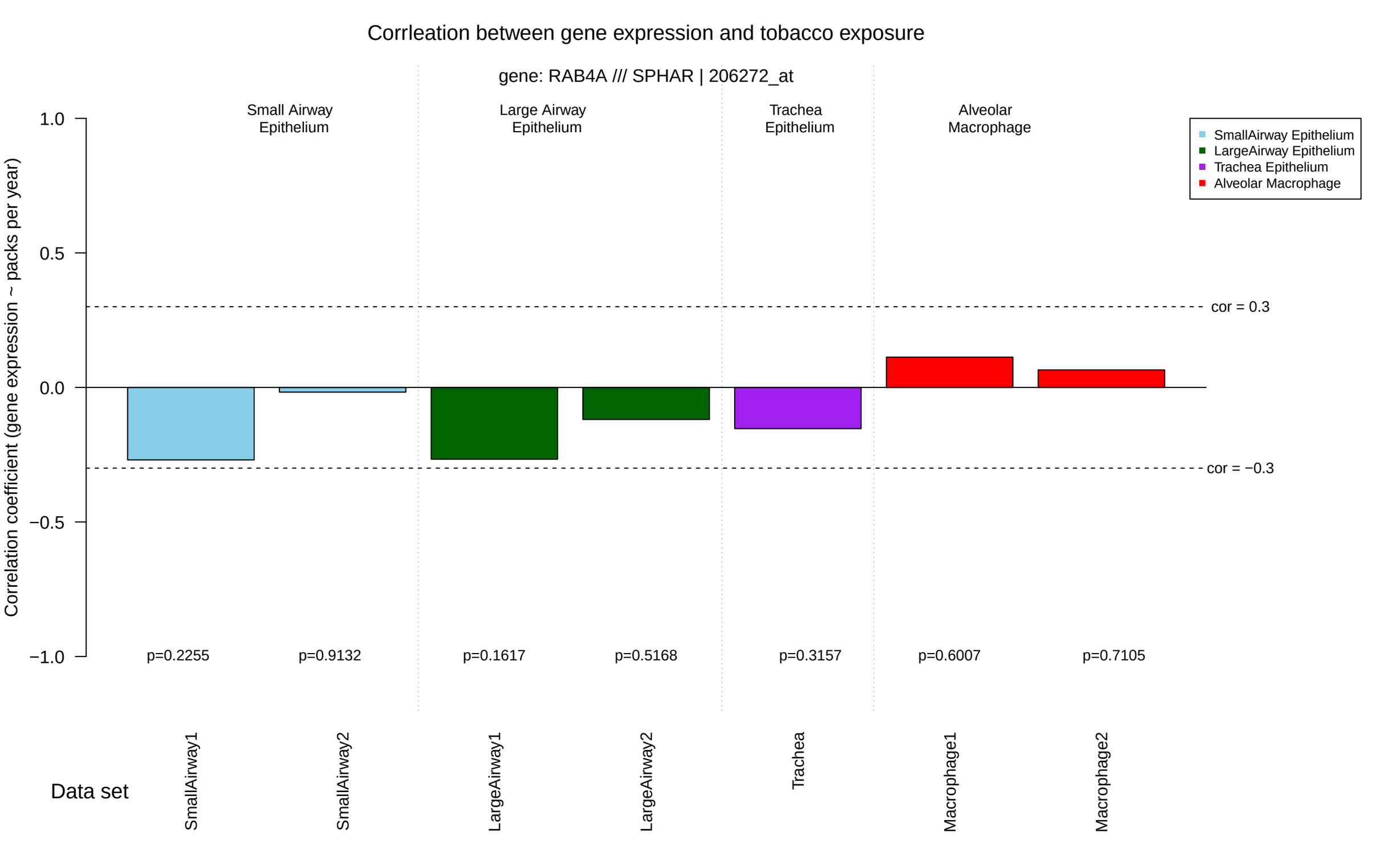

set: LargeAirway1 (p=0.0093)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0882)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0421)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7912)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9388)
RAB4A /// SPHAR | 206272_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB4A /// SPHAR | 206272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2255
p=0.9132
p=0.1617
p=0.5168
p=0.3157
p=0.6007
p=0.7105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2